

R Notebook

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v readr     2.1.5
## v ggplot2    4.0.0      v stringr  1.5.1
## v lubridate  1.9.4      v tibble   3.3.0
## v purrr      1.0.4      v tidyr    1.3.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(purrr)
library(ggplot2)
library(vegan)
```

```
## Loading required package: permute
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
```

```
library(stringr)
library(patchwork)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
knitr::opts_chunk$set(dev = "cairo_pdf")
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a",
                        full.names = TRUE)
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m")

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
pond_samples = c("s9", "s10", "s19", "s20")

# Create sample metadata using helper function
sample_metadata = create_sample_metadata(seq_metrics, pond_samples) %>%
  select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(experiment == 3) %>% # filter in specific experiments
  filter(water_control == "No") %>%
  filter(infection_class != "Definite") # filter out positive controls

species_data_filtered = all_data_species %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >= 3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)
```

```

# specific names that should group together
ent_v = "Enterovirus G"
circo_v = "Circovirus sp."
sapo_v = paste(c("Sapovirus", "Sapporo"), collapse = "|")
tes_v = "Teschovirus"
aich_v = "Aichivirus"
astr_v = paste(c("Porcine astrovirus", "Mamastrovirus 3", "Mamastrovirus 22", "Mamastrovirus pig", "PoAstV", "Pig astrovirus CX1"), collapse = "|")
hvp18 = "Alphapapillomavirus"
sapel_v = paste(c("Sapelovirus sp.", "Sapelovirus A"), collapse = "|")
boc_v = paste(c("Porcine bocavirus", "Bocaparvovirus", "Bocavirus pig"), collapse = "|")

# Virus pattern definitions
patterns <- c(
  EVG      = ent_v,
  PCV      = circo_v,
  PSapoV   = sapo_v,
  PTeV     = tes_v,
  PKV      = aich_v,
  PAstV    = astr_v,
  #HPV18   = hvp18,
  PSapeV   = sapel_v,
  PBoV     = boc_v
)

```

```

# Extract sample metadata (one row per sample)
sample_meta <- all_data_species %>%
  select(sample_name, library_prep, experiment, sample_type) %>%
  distinct()

# Create wide count matrix: taxa x samples
#count_mat <- species_data_filtered %>%
count_mat = all_data_species %>%
  group_by(sample_name, name) %>%
  summarise(reads = mean(nt_count, na.rm = TRUE), .groups = "drop") %>% # counting reads for each taxa
  pivot_wider(names_from = sample_name, values_from = reads, values_fill = 0) %>%
  column_to_rownames("name")

```

```

# ---- USER PARAMETERS ----
#depths <- c(5e5, 1e6, 2e6, 3e6, 4e6, 5e6, 7e6, 9e6, 11e6) # Target depths to simulate
depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6)
n_reps <- 1 # Replicates per depth

# Start the simulation!
res <- simulate_downsampling(count_mat, depths, n_reps)

```

```

# ---- SUMMARIZE RESULTS ----
# Diversity summary
div_summary <- res %>%
  group_by(depth) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),

```

```

    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By library prep
div_summary_libprep <- res %>%
  group_by(depth, library_prep) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By sample type
div_summary_sampletype <- res %>%
  group_by(depth, sample_type) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# remove _M from samples names for comparison
res = res %>%
  mutate(sample = str_remove(sample, "_M.*$"))

res_agg <- res %>%
  rename(richness_val = richness) %>%
  group_by(sample, library_prep, depth) %>%
  summarise(
    richness_mean = mean(richness_val),
    shannon_mean = mean(shannon),
    .groups = "drop"
  )

# do friedman test measuring richness by library prep pairing by sample
friedman_results <- res_agg %>%
  group_by(depth) %>%
  summarise(
    test = list(
      friedman.test(richness_mean ~ library_prep | sample)
    ),
    .groups = "drop"
  ) %>%
  mutate(
    statistic = map_dbl(test, ~ .$statistic),
    p_value = map_dbl(test, ~ .$p.value)
  )

```

```

# Join friedman results back to the overall table
div_summary = div_summary %>%
  left_join(friedman_results, by = "depth")

div_summary = div_summary %>%
  select(-shannon_sd, -richness_sd, -test, -statistic)

div_summary

```

```

## # A tibble: 15 x 4
##   depth richness_mean shannon_mean p_value
##   <dbl>         <dbl>         <dbl>   <dbl>
## 1 100000         4850.           5.87 0.00157
## 2 200000         6271.           5.89 0.00157
## 3 300000         7156.           5.90 0.00157
## 4 400000         7730.           5.90 0.00157
## 5 500000         8195.           5.90 0.00157
## 6 600000         8514.           5.90 0.0114
## 7 700000         8770.           5.91 0.0578
## 8 800000         8968.           5.91 0.0578
## 9 900000         9122.           5.91 0.206
##10 1000000        9255.           5.91 0.206
##11 1100000        9377.           5.91 0.206
##12 1200000        9466.           5.91 0.206
##13 1300000        9559.           5.91 0.206
##14 1400000        9635.           5.91 0.206
##15 1500000        9708.           5.91 0.206

```

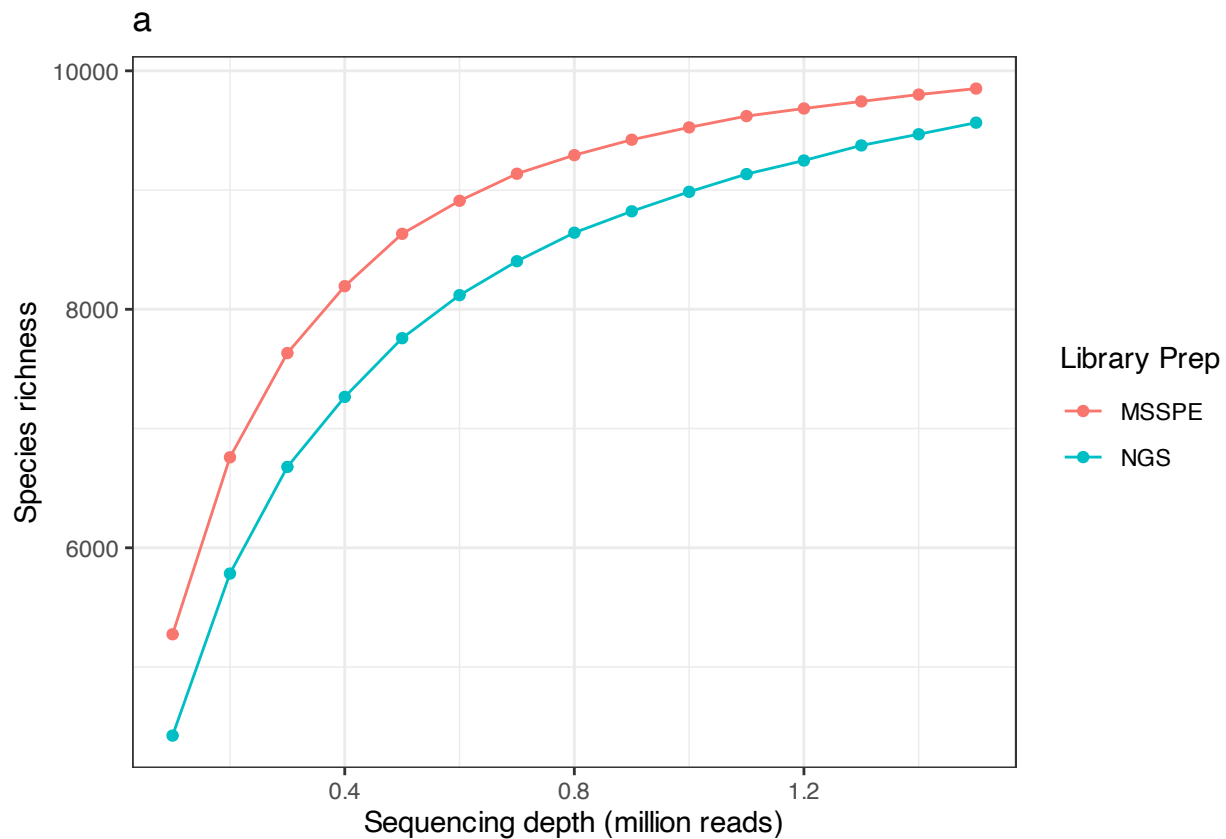
Rarefaction curves

```

div_summary_libprep_filter = div_summary_libprep %>%
  filter(depth <= 7.0e+06) %>%
  rename("Library Prep" = library_prep)

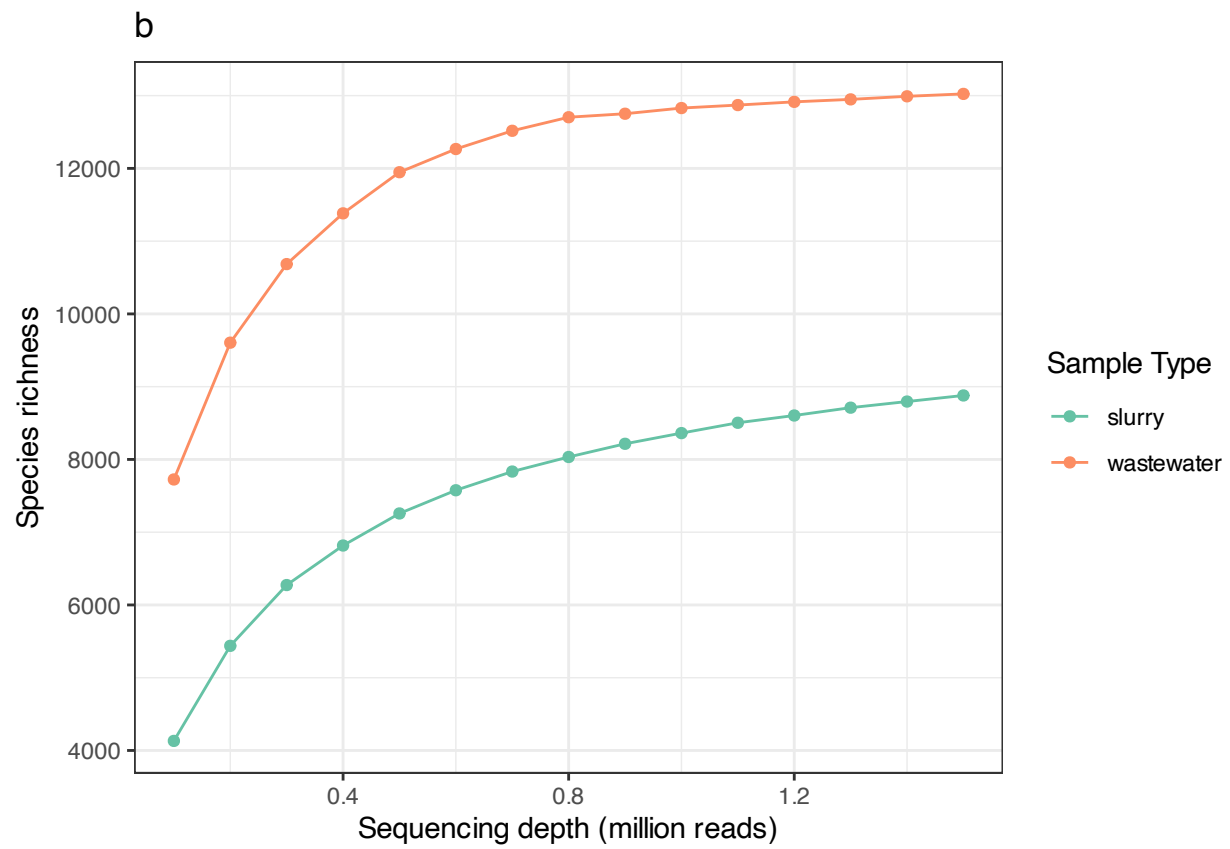
# Plot Rarefaction Curve
p1 <- ggplot(div_summary_libprep_filter, aes(x = depth/1e6, y = richness_mean, fill = `Library Prep`, c
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2)
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
    title = "a") +
  theme_bw()
p1

```

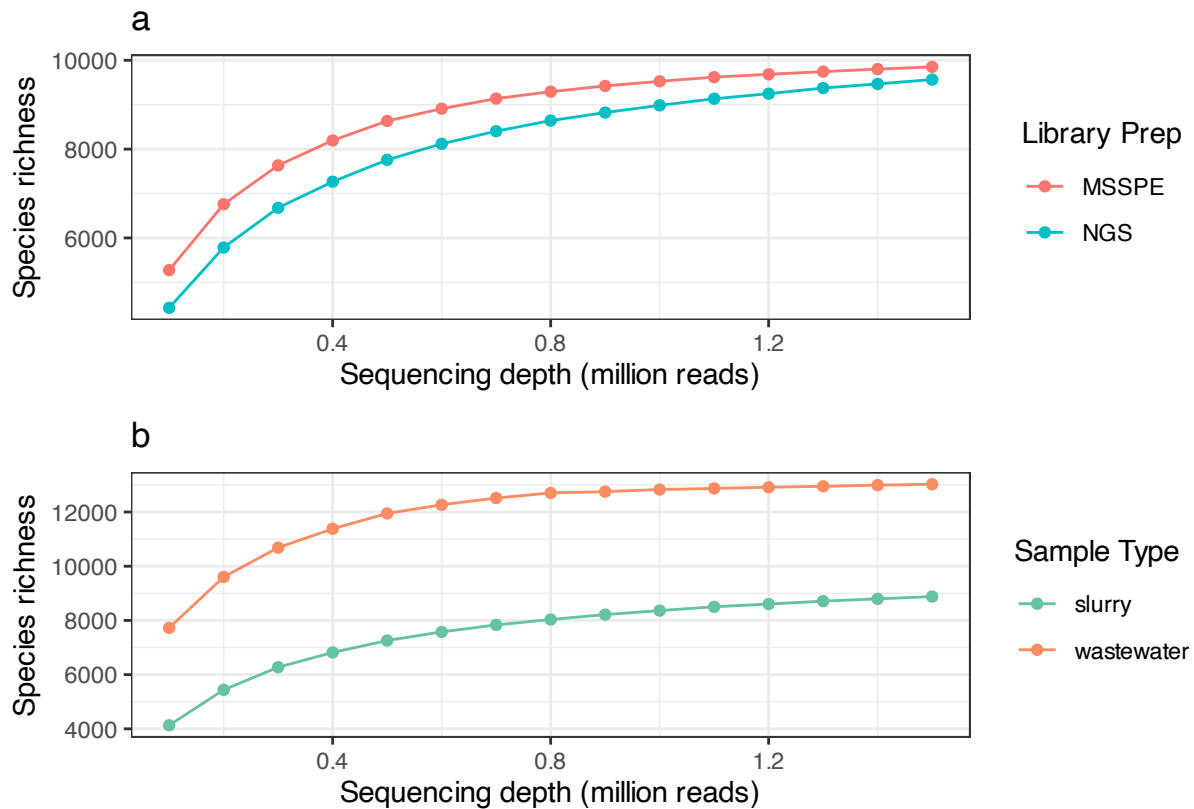


```
div_summary_sampletype_filter = div_summary_sampletype %>%
  filter(depth <= 7.0e+06) %>%
  mutate(
    sample_type = recode(sample_type, `environmental` = "wastewater")
  ) %>%
  rename("Sample Type" = sample_type)

# Plot Rarefaction Curve
p2 <- ggplot(div_summary_sampletype_filter, aes(x = depth/1e6, y = richness_mean, color = `Sample Type`)) +
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  scale_color_brewer(palette = "Set2") +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2)
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
    title = "b") +
  theme_bw()
p2
```



```
combined_curves = p1 / p2  
combined_curves
```



TARGET-SPECIFIC SEQ DEPTH ANALYSIS

```
# Start the simulation!
suppressWarnings(
  res_targets <- simulate_downsampling_targets(
    count_mat,
    depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6),
    n_reps = 10,
    patterns
  )
)

# Function to calculate detection probabilities from rarefied reads
summarize_detection <- function(res_targets, detection_threshold) {

  # 1. Pivot longer: each row = sample x replicate x target
  target_cols <- names(res_targets)[
    !(names(res_targets) %in% c("sample", "depth", "rep", "library_prep", "experiment", "sample_type"))
  ]

  det_long <- res_targets %>%
    pivot_longer(
      cols = all_of(target_cols),
      names_to = "target",
      values_to = "total_reads"
    ) %>%
    mutate(
      detected = as.integer(total_reads >= detection_threshold)
    )
}
```



```

)

# 2. Detection probability per depth and target
det_summary <- det_long %>%
  group_by(depth, target) %>%
  summarise(
    mean_total_reads = mean(total_reads, na.rm = TRUE),
    detect_prob = mean(detected, na.rm = TRUE),
    n_reps = n(),
    .groups = "drop"
  )

# 3. Detection probability by library prep
det_summary_libprep <- det_long %>%
  group_by(depth, library_prep, target) %>%
  summarise(
    mean_total_reads = mean(total_reads, na.rm = TRUE),
    detect_prob = mean(detected, na.rm = TRUE),
    n_reps = n(),
    .groups = "drop"
  )

list(
  long = det_long,
  summary = det_summary,
  summary_libprep = det_summary_libprep
)
}

# Summarized detection probabilities from simulation
res_summary <- summarize_detection(res_targets, detection_threshold = 3)

# Access outputs
det_long <- res_summary$long # replicate-level total reads + detected 0/1
det_summary <- res_summary$summary # mean reads + detection prob per depth x target
det_summary_libprep <- res_summary$summary_libprep # mean reads + detection prob per depth x target x

```

Linear Mixed Model for Diversity

```
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.17
## Current TMB package version is 1.9.18
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more)

```

```
library(performance)
```

```

# get response data from simulation
model_div_data = res %>%
  mutate(
    depth_scaled = depth / 1e5, # scale depth

```

```

library_prep = factor(library_prep), # convert to factor
library_prep = relevel(library_prep, ref = "NGS") # set NGS as reference
)

# create base model for diversity metrics
model_seq_depth_div <- glmmTMB(
  richness ~ library_prep * depth_scaled * sample_type + (1 | sample),
  data = model_div_data,
  family = nbinom2() # use if Richness
)

# print summary of model performance
summary(model_seq_depth_div)

```

```

## Family: nbinom2 ( log )
## Formula:
## richness ~ library_prep * depth_scaled * sample_type + (1 | sample)
## Data: model_div_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    5025.8    5062.9   -2502.9    5005.8      290
##
## Random effects:
##
## Conditional model:
##   Groups Name      Variance Std.Dev.
## sample (Intercept) 0.04559  0.2135
## Number of obs: 300, groups: sample, 10
##
## Dispersion parameter for nbinom2 family (): 72.1
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)      9.083341   0.158032  57.48
## library_prepMSSPE    0.183474   0.065870   2.79
## depth_scaled        0.031744   0.005170   6.14
## sample_typeslurry   -0.607472   0.176708  -3.44
## library_prepMSSPE:depth_scaled -0.012074   0.007293  -1.66
## library_prepMSSPE:sample_typeslurry -0.041153   0.073764  -0.56
## depth_scaled:sample_typeslurry    0.014500   0.005790   2.50
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.004555   0.008169   0.56
##
##              Pr(>|z|)
## (Intercept)      < 2e-16 ***
## library_prepMSSPE    0.005346 **
## depth_scaled        8.26e-10 ***
## sample_typeslurry   0.000587 ***
## library_prepMSSPE:depth_scaled    0.097802 .
## library_prepMSSPE:sample_typeslurry 0.576913
## depth_scaled:sample_typeslurry    0.012277 *
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.577097
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth_div, parm="beta_", method="profile"))
```

```
##                                2.5 %      97.5 %
## (Intercept)                   6282.5673643 1.234719e+04
## library_prepMSSPE             1.0553022 1.367555e+00
## depth_scaled                  1.0218046 1.042805e+00
## sample_typeslurry             0.3733389 7.946939e-01
## library_prepMSSPE:depth_scaled 0.9739179 1.002270e+00
## library_prepMSSPE:sample_typeslurry 0.8299907 1.109500e+00
## depth_scaled:sample_typeslurry 1.0031063 1.026224e+00
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.9885450 1.020834e+00
```

```
# overdispersion
overdisp_fun(model_seq_depth_div)
```

```
##      chisq      ratio      rdf      p
## 265.4262575  0.9152630 290.0000000 0.8468029
```

Microbial diversity sequencing depth tables

```
cat("\\captionsetup{labelformat=empty}")
```

```
# Fixed effects table
```

```
fixed_tbl_div <- summary(model_seq_depth_div)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)
```

```
# Add odds ratio and 95% CI
```

```
fixed_tbl_div <- fixed_tbl_div %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Percent_change = (OR - 1) * 100,
    Effect = recode(Effect,
      `(Intercept)` = "intercept",
      `library_prepMSSPE` = "Library prep",
      `depth_scaled` = "Sequencing depth",
      `sample_typeslurry` = "Sample type",
      `library_prepMSSPE:depth_scaled` = "Lib prep * Sequencing depth",
      `library_prepMSSPE:sample_typeslurry` = "Lib prep * Sample type",
      `depth_scaled:sample_typeslurry` = "Sequencing depth * Sample type",
      `library_prepMSSPE:depth_scaled:sample_typeslurry` = "Lib Prep * Depth * Sample Type"
    )
  ) %>%
  rename(
    "Percent change" = Percent_change,
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )
```

```

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth_div)

random_tbl_div <- data.frame(
  Random_Effect = c("Virus target", "Sample"),
  Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
  SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl_div, Random_Effects = random_tbl_div)

```

```

$Fixed_Effects Effect Estimate SE z 1 intercept 9.083340513 0.158031851 57.4779099 2 Library prep
0.183474243 0.065869808 2.7854073 3 Sequencing depth 0.031743583 0.005170130 6.1398039 4 Sample type
-0.607471568 0.176708488 -3.4377045 5 Lib prep * Sequencing depth -0.012073962 0.007292794 -1.6556018 6
Lib prep * Sample type -0.041153120 0.073764354 -0.5578998 7 Sequencing depth * Sample type 0.014499809
0.005790447 2.5040915 8 Lib Prep * Depth * Sample Type 0.004555112 0.008168705 0.5576296 p OR CI
25% CI 75% Percent change 1 0.000000e+00 8807.3379714 6461.3792790 1.200505e+04 8.806338e+05
2 5.346053e-03 1.2013840 1.0558746 1.366946e+00 2.013840e+01 3 8.262343e-10 1.0322528 1.0218453
1.042766e+00 3.225278e+00 4 5.866675e-04 0.5447264 0.3852664 7.701862e-01 -4.552736e+01 5 9.780248e-
02 0.9879986 0.9739768 1.002222e+00 -1.200136e+00 6 5.769128e-01 0.9596822 0.8304968 1.108963e+00
-4.031783e+00 7 1.227663e-02 1.0146054 1.0031555 1.026186e+00 1.460544e+00 8 5.770973e-01 1.0045655
0.9886098 1.020779e+00 4.565502e-01

```

```

$Random_Effects Random_Effect Variance SD 1 Virus target 0.04559006 0.2135183 2 Sample 0.04559006
0.2135183

```

```

# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl_div, digits = 3, caption = "Fixed effects from GLMM (negative binomial)")

```

Fixed effects from GLMM (negative binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%	Percent change
intercept	9.083	0.158	57.478	0.000	8807.338	6461.379	12005.053	880633.797
Library prep	0.183	0.066	2.785	0.005	1.201	1.056	1.367	20.138
Sequencing depth	0.032	0.005	6.140	0.000	1.032	1.022	1.043	3.225
Sample type	-0.607	0.177	-3.438	0.001	0.545	0.385	0.770	-45.527
Lib prep * Sequencing depth	-0.012	0.007	-1.656	0.098	0.988	0.974	1.002	-1.200
Lib prep * Sample type	-0.041	0.074	-0.558	0.577	0.960	0.830	1.109	-4.032
Sequencing depth * Sample type	0.014	0.006	2.504	0.012	1.015	1.003	1.026	1.461
Lib Prep * Depth * Sample Type	0.005	0.008	0.558	0.577	1.005	0.989	1.021	0.457

```

kable(random_tbl_div, digits = 3, caption = "Random effects (variance components) from GLMM")

```

Random effects (variance components) from GLMM

Random_Effect	Variance	SD
Virus target	0.046	0.214
Sample	0.046	0.214

Linear Mixed Model for Targets

```
library(glmmTMB)
library(broom.mixed)
library(performance)

det_long <- det_long %>%
  mutate(
    depth_scaled = depth / 1e5,          # scale depth
    library_prep = factor(library_prep), # convert to factor
    library_prep = relevel(library_prep, ref = "NGS") # set NGS as reference
  )

# create base model for detection probability
model_seq_depth <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1 | target) + (1 | sample),
  data = det_long,
  family = binomial(link = "logit")
)

# print summary of model performance
summary(model_seq_depth)
```

```
## Family: binomial ( logit )
## Formula:      detected ~ library_prep * depth_scaled + (1 | target) + (1 |
##      sample)
## Data: det_long
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##   8476.3   8524.8  -4232.2   8464.3    23994
##
## Random effects:
##
## Conditional model:
## Groups Name      Variance Std.Dev.
## target (Intercept) 22.604   4.754
## sample (Intercept)  8.896   2.983
## Number of obs: 24000, groups: target, 8; sample, 20
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.01647    1.95520   1.031   0.302
## library_prepMSSPE 2.19810    1.34009   1.640   0.101
## depth_scaled      0.21180    0.01010  20.979 < 2e-16 ***
## library_prepMSSPE:depth_scaled -0.08629    0.01348  -6.399 1.56e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth, parm="beta_", method="profile"))
```

```
##              2.5 %      97.5 %
## (Intercept)    0.1238410 630.0661386
## library_prepMSSPE 0.5705955 143.2210859
## depth_scaled    1.2118241  1.2607667
## library_prepMSSPE:depth_scaled 0.8933406  0.9418527
```

```
# standard error
summary(model_seq_depth)$coefficients$cond
```

```
##              Estimate Std. Error  z value    Pr(>|z|)
## (Intercept)    2.01647044 1.95519644  1.031339 3.023819e-01
## library_prepMSSPE 2.19810151 1.34008562  1.640269 1.009492e-01
## depth_scaled    0.21179917 0.01009576 20.979024 1.019657e-97
## library_prepMSSPE:depth_scaled -0.08628557 0.01348323 -6.399474 1.559132e-10
## attr(,"ddf")
## [1] "asymptotic"
```

```
# Extract coefficients
```

```
coefs <- summary(model_seq_depth)$coefficients$cond
```

```
# Compute OR, CI, p-value, and % change
```

```
or_table <- data.frame(
  term = rownames(coefs),
  estimate = coefs[, "Estimate"],
  std_error = coefs[, "Std. Error"],
  z_value = coefs[, "z value"],
  p_value = coefs[, "Pr(>|z|)"],
  OR = exp(coefs[, "Estimate"]),
  CI_low = exp(coefs[, "Estimate"] - 1.96 * coefs[, "Std. Error"]),
  CI_high = exp(coefs[, "Estimate"] + 1.96 * coefs[, "Std. Error"])
) |>
  dplyr::mutate(
    percent_change = (OR - 1) * 100 # negative = reduction, positive = increase
  )
```

```
or_table
```

```
##              term      estimate
## (Intercept)    (Intercept) 2.01647044
## library_prepMSSPE library_prepMSSPE 2.19810151
## depth_scaled    depth_scaled 0.21179917
## library_prepMSSPE:depth_scaled library_prepMSSPE:depth_scaled -0.08628557
##              std_error  z_value    p_value      OR
## (Intercept)    1.95519644  1.031339 3.023819e-01 7.5117649
## library_prepMSSPE 1.34008562  1.640269 1.009492e-01 9.0078958
## depth_scaled    0.01009576 20.979024 1.019657e-97 1.2358997
## library_prepMSSPE:depth_scaled 0.01348323 -6.399474 1.559132e-10 0.9173322
##              CI_low    CI_high percent_change
## (Intercept)    0.1627216 346.7678275      651.176490
```

```
## library_prepMSSPE          0.6515075 124.5452771      800.789584
## depth_scaled              1.2116844   1.2605988       23.589966
## library_prepMSSPE:depth_scaled 0.8934073   0.9418979       -8.266777
```

```
# check residuals
```

```
r2_model <- r2_nakagawa(model_seq_depth)
```

```
r2_model
```

```
## # R2 for Mixed Models
```

```
##
```

```
## Conditional R2: 0.908
```

```
## Marginal R2: 0.032
```

```
# check overdispersion
```

```
overdisp_fun(model_seq_depth)
```

```
##          chisq          ratio          rdf          p
## 42999.617763      1.792099 23994.000000      0.000000
```

```
# Fit zero-inflated model for comparison
```

```
model_zi <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1|target) + (1|sample),
  ziformula = ~1,
  data = det_long,
  family = binomial
)
```

```
# Compare AIC
```

```
AIC(model_seq_depth, model_zi)
```

```
##          df          AIC
## model_seq_depth  6 8476.333
## model_zi         7 8775.062
```

```
# Compare alternative sequencing depth modeling
```

```
# try log transformed depth
```

```
det_long <- det_long %>%
  mutate(log_depth = log(depth))
```

```
model_log <- glmmTMB(
  detected ~ library_prep * log_depth + (1|target) + (1|sample),
  data = det_long,
  family = binomial
)
```

```
# print AIC comparison
```

```
AIC(model_seq_depth, model_log)
```

```
##          df          AIC
## model_seq_depth  6 8476.333
## model_log        6 8142.894
```

Viral target sequencing depth tables

```
cat("\\captionsetup{labelformat=empty}")

# Fixed effects table
fixed_tbl <- summary(model_seq_depth)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)

# Add odds ratio and 95% CI
fixed_tbl <- fixed_tbl %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Effect = recode(Effect,
      `(Intercept)` = "intercept",
      `library_prepMSSPE` = "Library prep",
      `depth_scaled` = "Sequencing depth",
      `library_prepMSSPE:depth_scaled` = "Library prep * Sequencing depth"
    )
  ) %>%
  rename(
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth)

random_tbl <- data.frame(
  Random_Effect = c("Virus target", "Sample"),
  Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
  SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl, Random_Effects = random_tbl)
```

```
$Fixed_Effects Effect Estimate SE z p 1 intercept 2.01647044 1.95519644 1.031339 3.023819e-01 2 Library
prep 2.19810151 1.34008562 1.640269 1.009492e-01 3 Sequencing depth 0.21179917 0.01009576 20.979024
1.019657e-97 4 Library prep * Sequencing depth -0.08628557 0.01348323 -6.399474 1.559132e-10 OR CI
25% CI 75% 1 7.5117649 0.1627216 346.7678275 2 9.0078958 0.6515075 124.5452771 3 1.2358997 1.2116844
1.2605988 4 0.9173322 0.8934073 0.9418979
```

```
$Random_Effects Random_Effect Variance SD 1 Virus target 22.604090 4.754376 2 Sample 8.895867
2.982594
```

```
# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl, digits = 3, caption = "Fixed effects from GLMM (binomial)")
```


Fixed effects from GLMM (binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%
intercept	2.016	1.955	1.031	0.302	7.512	0.163	346.768
Library prep	2.198	1.340	1.640	0.101	9.008	0.652	124.545
Sequencing depth	0.212	0.010	20.979	0.000	1.236	1.212	1.261
Library prep * Sequencing depth	-0.086	0.013	-6.399	0.000	0.917	0.893	0.942

```
kable(random_tbl, digits = 3, caption = "Random effects (variance components) from GLMM")
```

Random effects (variance components) from GLMM

Random_Effect	Variance	SD
Virus target	22.604	4.754
Sample	8.896	2.983

Viral sequencing depth plots

```
# Create a grid of sequencing depths for prediction
depth_grid <- seq(min(det_long$depth_scaled),
                  max(det_long$depth_scaled),
                  length.out = 100)

newdata <- expand.grid(
  library_prep = c("MSSPE", "NGS"),
  depth_scaled = depth_grid
)

# Predicted probabilities and 95% CI using emmeans
emm <- emmeans(model_seq_depth, ~ library_prep | depth_scaled,
               at = list(depth_scaled = depth_grid), type="response")

emm_df <- as.data.frame(emm)

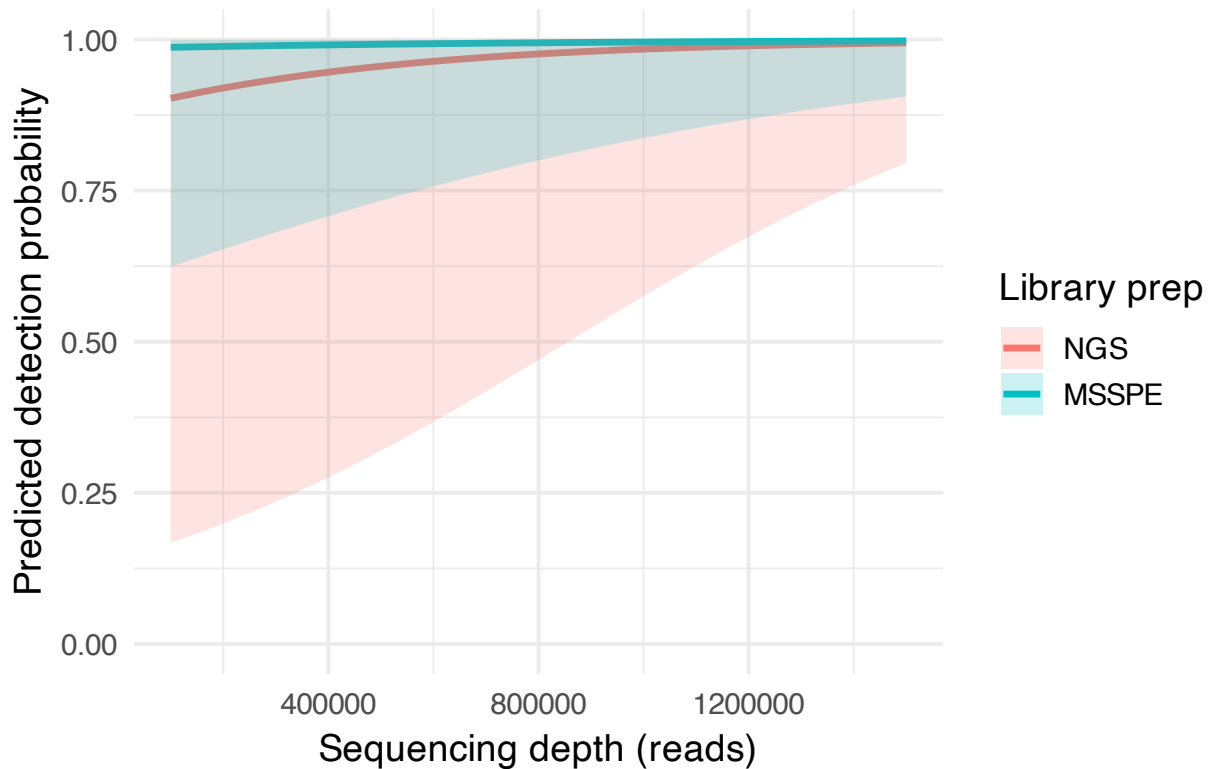
# Rename columns for clarity
emm_df <- emm_df %>%
  rename(Detection_Prob = prob,
         CI_Lower = asymp.LCL,
         CI_Upper = asymp.UCL)

# Plot
ggplot(emm_df, aes(x = depth_scaled*1e5, y = Detection_Prob, color = library_prep)) +
  geom_line(size = 1.2) +
  geom_ribbon(aes(ymin = CI_Lower, ymax = CI_Upper, fill = library_prep), alpha = 0.2, color = NA) +
  scale_x_continuous(name = "Sequencing depth (reads)") +
  scale_y_continuous(name = "Predicted detection probability", limits = c(0,1)) +
  labs(color = "Library prep", fill = "Library prep") +
  theme_minimal(base_size = 14) +
  ggtitle("Predicted virus detection probability vs. sequencing depth")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Predicted virus detection probability vs. sequencing depth



Tables

```
cat("\captionsetup{labelformat=empty}")

det_summary_libprep_box_format <- det_summary_libprep %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
            .groups = "drop")

det_summary_libprep_box_format
```

A tibble: 240 x 4

```
depth library_prep target mean_prob
<dbl> <chr> <chr> <dbl>
1 100000 MSSPE EVG 0.66 2 100000 MSSPE PAsV 1
3 100000 MSSPE PBoV 1
4 100000 MSSPE PCV 0.66 5 100000 MSSPE PKV 0.03 6 100000 MSSPE PSapeV 0.53 7 100000 MSSPE
PSapoV 0.78 8 100000 MSSPE PTeV 0.58 9 100000 NGS EVG 0.49 10 100000 NGS PAsV 1
# i 230 more rows
```

```

det_summary_wide <- det_summary_libprep_box_format %>%
  group_by(target, depth, library_prep) %>% # average across library preps if multiple
  summarise(mean_prob = mean(mean_prob), .groups = "drop") %>%
  mutate(
    depth = paste0(depth / 1e6, "M")
  ) %>%
  pivot_wider(
    names_from = c(depth),
    values_from = mean_prob
  )

det_summary_wide

```

A tibble: 16 x 17

```

target library_prep 0.1M 0.2M 0.3M 0.4M 0.5M 0.6M 0.7M 0.8M 1 EVG MSSPE 0.66 0.78 0.86 0.89 0.9 0.9
0.9 0.9 2 EVG NGS 0.49 0.64 0.75 0.74 0.84 0.87 0.88 0.88 3 PAsTV MSSPE 1 1 1 1 1 1 1 1
4 PAsTV NGS 1 1 1 1 1 1 1 1
5 PBoV MSSPE 1 1 1 1 1 1 1 1
6 PBoV NGS 0.92 0.99 1 0.99 1 1 1 1
7 PCV MSSPE 0.66 0.8 0.9 0.9 0.9 0.9 0.9 0.9 8 PCV NGS 0.49 0.64 0.71 0.74 0.75 0.76 0.78 0.79 9 PKV
MSSPE 0.03 0.13 0.23 0.28 0.29 0.29 0.3 0.31 10 PKV NGS 0 0.01 0.02 0.01 0.04 0.03 0.07 0.09 11 PSapeV
MSSPE 0.53 0.6 0.6 0.6 0.6 0.6 0.6 0.6 12 PSapeV NGS 0.4 0.6 0.71 0.71 0.75 0.72 0.77 0.78 13 PSapoV
MSSPE 0.78 0.95 0.99 1 1 1 1 1
14 PSapoV NGS 0.7 0.89 0.97 0.98 0.99 1 1 1
15 PTeV MSSPE 0.58 0.63 0.7 0.69 0.74 0.78 0.79 0.8 16 PTeV NGS 0.48 0.59 0.6 0.65 0.67 0.68 0.69 0.7 #
i 7 more variables: 0.9M , 1M , 1.1M , 1.2M , # 1.3M , 1.4M , 1.5M

```

```

det_summary_wide %>%
  rename(
    "Virus" = target,
    "Library Prep" = library_prep
  ) %>%
  select(`0.2M`, `0.4M`, `0.6M`, `0.8M`, `1M`, `1.2M`, `1.4M`) %>%
  kable(
    format = "latex", # or "latex" if using LaTeX
    caption = "Detection probability per viral target at varying sequencing depths",
    digits = 2,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )

```

```

#Threshold 1
res_summary <- summarize_detection(res_targets, detection_threshold = 1)
det_long <- res_summary$long # replicate-level total reads + detected 0/1
det_summary <- res_summary$summary # mean reads + detection prob per depth x target
det_summary_libprep <- res_summary$summary_libprep # mean reads + detection prob per depth x target x

```

Detection probability per viral target at varying sequencing depths

Virus	Library Prep	0.1M	0.3M	0.5M	0.7M	0.9M	1.1M	1.3M	1.5M
EVG	MSSPE	0.66	0.86	0.90	0.90	0.90	0.90	0.9	0.90
EVG	NGS	0.49	0.75	0.84	0.88	0.90	0.90	0.9	0.90
PAstV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.0	1.00
PAstV	NGS	1.00	1.00	1.00	1.00	1.00	1.00	1.0	1.00
PBoV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.0	1.00
PBoV	NGS	0.92	1.00	1.00	1.00	1.00	1.00	1.0	1.00
PCV	MSSPE	0.66	0.90	0.90	0.90	0.90	0.90	0.9	0.90
PCV	NGS	0.49	0.71	0.75	0.78	0.80	0.79	0.8	0.80
PKV	MSSPE	0.03	0.23	0.29	0.30	0.29	0.32	0.3	0.31
PKV	NGS	0.00	0.02	0.04	0.07	0.07	0.10	0.1	0.10
PSapeV	MSSPE	0.53	0.60	0.60	0.60	0.60	0.60	0.6	0.60
PSapeV	NGS	0.40	0.71	0.75	0.77	0.80	0.80	0.8	0.80
PSapoV	MSSPE	0.78	0.99	1.00	1.00	1.00	1.00	1.0	1.00
PSapoV	NGS	0.70	0.97	0.99	1.00	1.00	1.00	1.0	1.00
PTeV	MSSPE	0.58	0.70	0.74	0.79	0.80	0.80	0.8	0.80
PTeV	NGS	0.48	0.60	0.67	0.69	0.70	0.70	0.7	0.70

```

det_summary_libprep_box_format <- det_summary_libprep %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
    .groups = "drop")

# Threshold 2
res_summary_2 <- summarize_detection(res_targets, detection_threshold = 5)
det_long_2 <- res_summary_2$long # replicate-level total reads + detected 0/1
det_summary_2 <- res_summary_2$summary # mean reads + detection prob per depth x target
det_summary_libprep_2 <- res_summary_2$summary_libprep # mean reads + detection prob per depth x target

det_summary_libprep_box_format_2 <- det_summary_libprep_2 %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
    .groups = "drop")

#Threshold 3
res_summary_3 <- summarize_detection(res_targets, detection_threshold = 10)
det_long_3 <- res_summary_3$long # replicate-level total reads + detected 0/1
det_summary_3 <- res_summary_3$summary # mean reads + detection prob per depth x target
det_summary_libprep_3 <- res_summary_3$summary_libprep # mean reads + detection prob per depth x target

det_summary_libprep_box_format_3 <- det_summary_libprep_3 %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
    .groups = "drop")

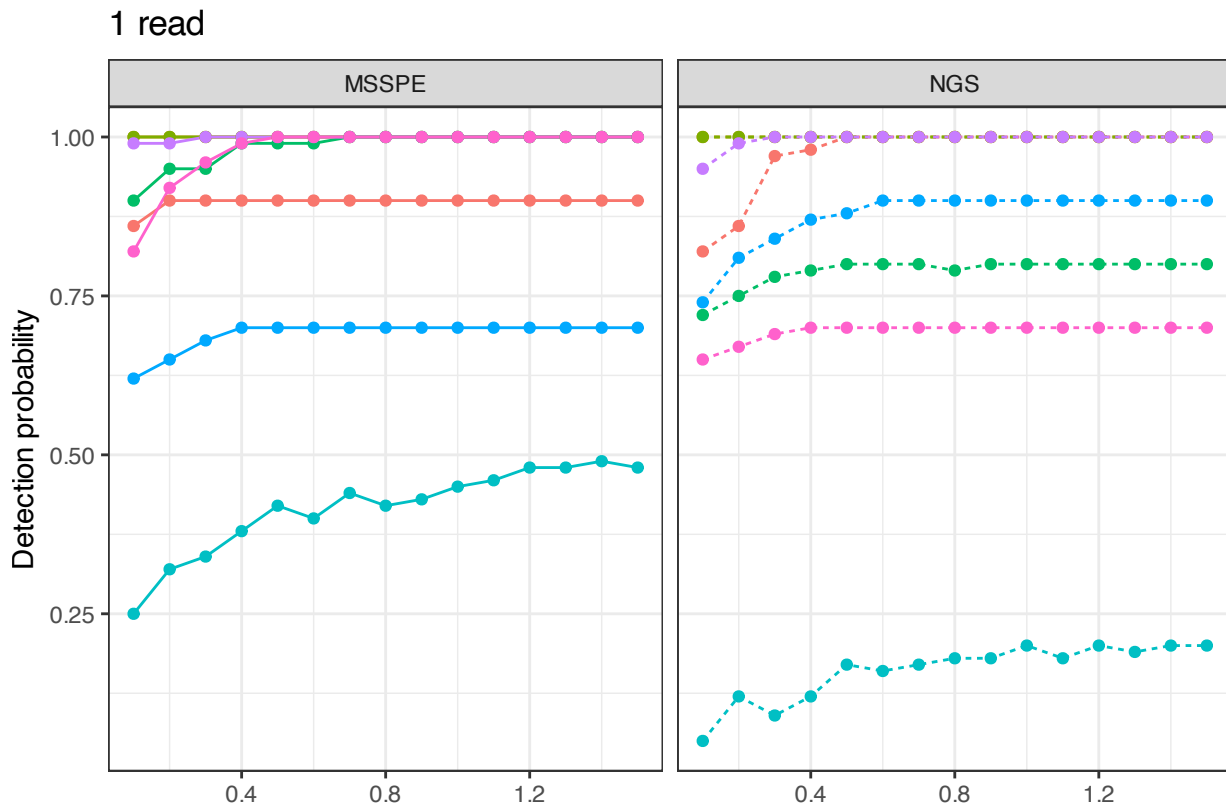
# Plot Library Prep Curve
rare_targets_curve_prep <- ggplot(det_summary_libprep_box_format, aes(x = depth/1e6, y = mean_prob, col = library_prep)) +
  geom_line() + # small horizontal dodge to separate lines

```

```

#geom_point(position = position_jitter(width = 0.3, height = 0.0)) + # jitter x slightly
geom_point() +
facet_wrap(~library_prep) +
labs(x = "", y = "Detection probability",
      title = "1 read") +
theme_bw() +
theme(
  legend.position = "none",
)
rare_targets_curve_prep

```

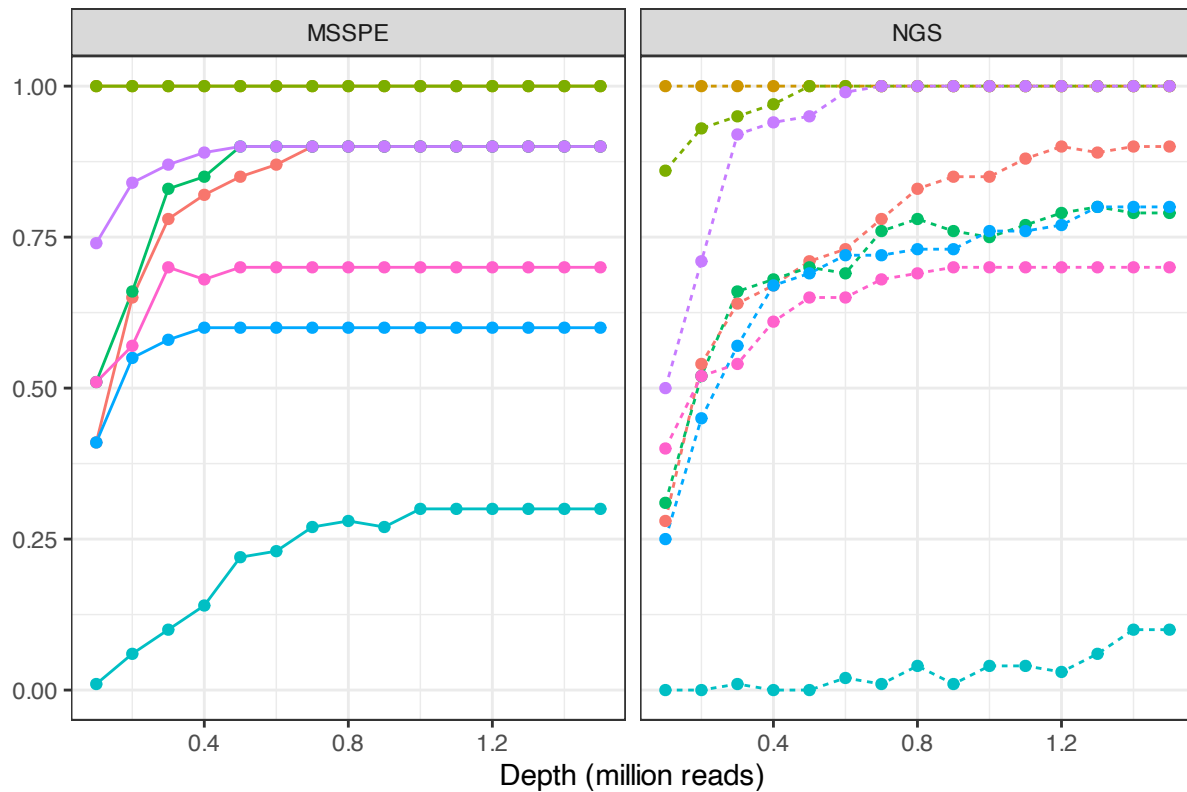


```

# Plot Library Prep Curve 2nd Depth
rare_targets_curve_prep_2 <- ggplot(det_summary_libprep_box_format_2, aes(x = depth/1e6, y = mean_prob,
  geom_line() + # small horizontal dodge to separate lines
  #geom_point(position = position_jitter(width = 0.3, height = 0.0)) + # jitter x slightly
  geom_point() +
  facet_wrap(~library_prep) +
  labs(x = "Depth (million reads)", y = "",
        title = "5 reads") +
  theme_bw() +
  theme(
    legend.position = "none",
  )
rare_targets_curve_prep_2

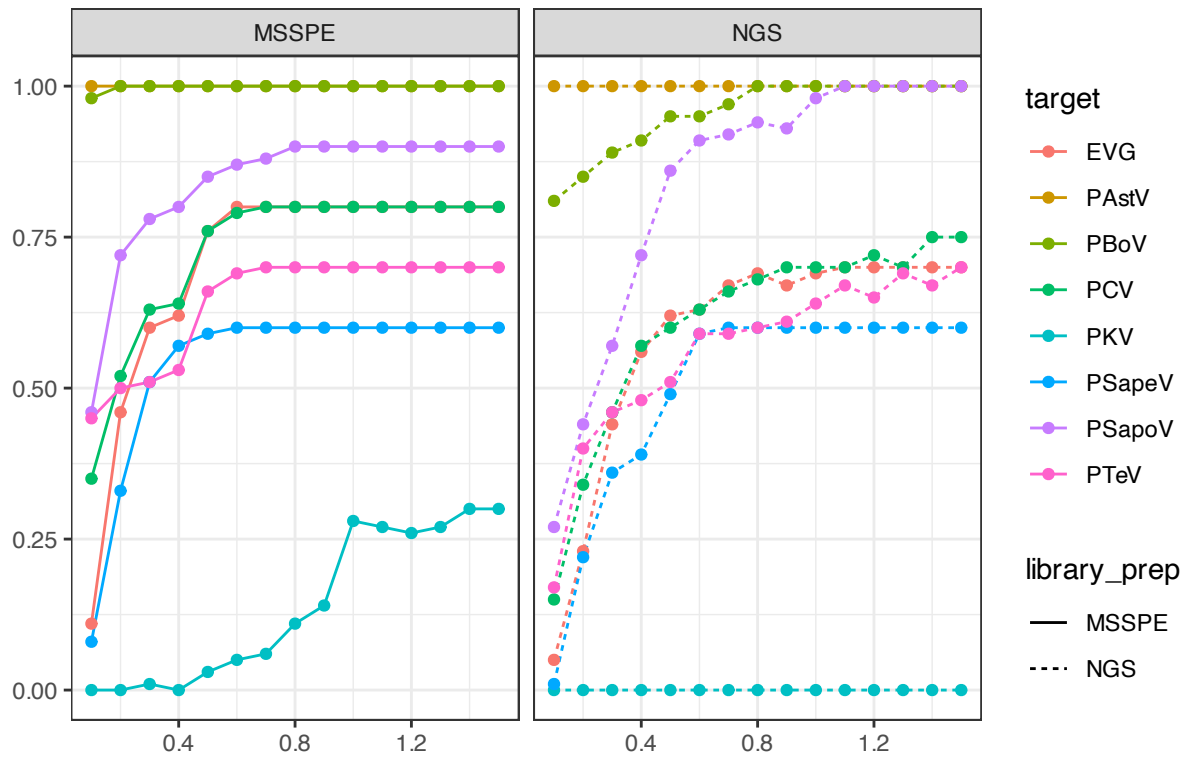
```

5 reads



```
# Plot Library Prep Curve 3rd Depth
rare_targets_curve_prep_3 <- ggplot(det_summary_libprep_box_format_3, aes(x = depth/1e6, y = mean_prob,
  geom_line() + # small horizontal dodge to separate lines
  #geom_point(position = position_jitter(width = 0.3, height = 0.0)) + # jitter x slightly
  geom_point() +
  facet_wrap(~library_prep) +
  labs(x = "", y = "",
    title = "10 reads") +
  theme_bw())
rare_targets_curve_prep_3
```

10 reads



```
combined_target_curves = rare_targets_curve_prep | rare_targets_curve_prep_2 | rare_targets_curve_prep_3
plot_layout(ncol = 3, widths = c(1, 1, 1), guides = "collect") &
theme(plot.margin = unit(c(10.5, 10.5, 10.5, 10.5), "cm"))
```

```
## NULL
```

```
combined_target_curves
```

